OSY-W-23 - OSY 22516 winter 2023 model answer paper. OSY model answer paper winter 2023.

Computer Engineering (Anjuman-I-Islam’s A. R. Kalsekar Polytechnic)

WINTER – 2023 EXAMINATION
MODEL ANSWER

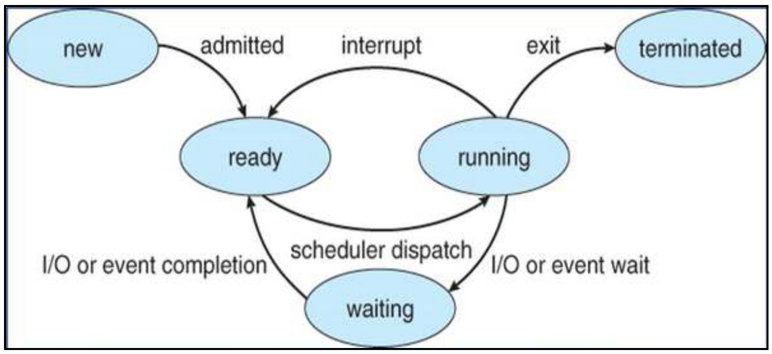
Subject: Operating Systems **Subject Code:** 22516

Important Instructions to examiners:

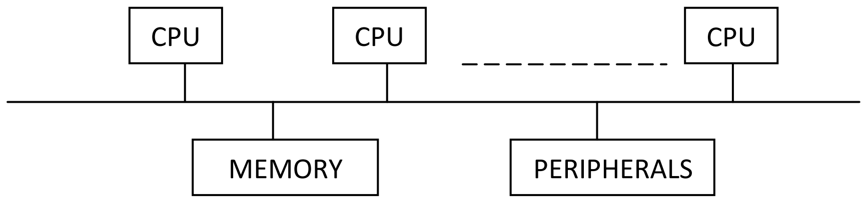
1) The answers should be examined by key words and not as word-to-word as given in the model answer scheme.
2) The model answer and the answer written by candidate may vary but the examiner may try to assess the understanding level of the candidate.
3) The language errors such as grammatical, spelling errors should not be given more Importance (Not applicable for subject English and Communication Skills.
4) While assessing figures, examiner may give credit for principal components indicated in the figure. The figures drawn by candidate and model answer may vary. The examiner may give credit for anyequivalent figure drawn.
5) Credits may be given step wise for numerical problems. In some cases, the assumed constant values may vary and there may be some difference in the candidate's answers and model answer.
6) In case of some questions credit may be given by judgement on part of examiner of relevant answer based on candidate's understanding.
7) For programming language papers, credit may be given to any other program based on equivalent concept.
8) As per the policy decision of Maharashtra State Government, teaching in English/Marathi and Bilingual (English + Marathi) medium is introduced at first year of AICTE diploma Programme from academic year 2021-2022. Hence if the students in first year (first and second semesters) write answers in Marathi or bilingual language (English +Marathi), the Examiner shall consider the same and assess the answer based on matching of concepts with model answer.

Q. No	Sub Q.N.	Answer	Marking Scheme
1.		**Attempt any <u>FIVE</u> of the following:**	**10**
	a)	**Define real time operating system, along with any two applications of it.**	**2M**
	Ans.	Real Time operating system is a special-purpose operating system used in computers that has well defined fixed time constraints for processing any task. **OR** Real-time operating systems (RTOS) are used in environments where a large number of events, mostly external to the computer system, must be accepted and processed in a short time or within certain deadlines. **OR Any relevant Definition** Applications: (any other relevant application shall be considered)	***Definition 1M, Any two applications 1/2M each***

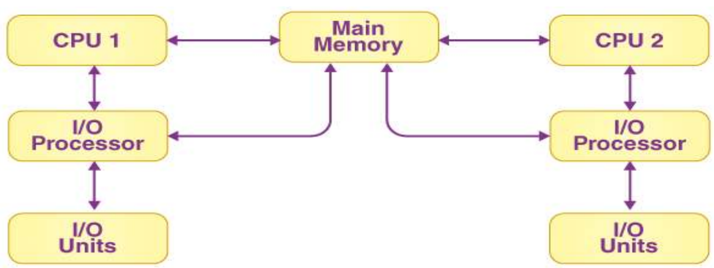
		1. Flight Control System 2. Simulations 3. Industrial control 4. Military applications 5. Airline Traffic Control System 6. Airline Reservation System 7. Defence application systems like RADAR 8. Heart Pacemaker 9. Network Multimedia Systems 10. Missile Guidance system 11. Weather Forecasting 12. Online Transaction system 13. Network and Multimedia Systems 14. Ticket Reservation System 15. Command Control Systems 16. Medical Critical Care Systems.	
	b) **Ans.**	**List any four services provided by operating system.** • User Interface • Program Execution • I/O Operation • File system Manipulation • Communication • Error Detection • Resource Allocation • Accounting • Protection and security	**2M** ***Any four 1/2M each***
	c)	**Draw neat labelled process state diagram along with the correct directions of arrows**	**2M**

	Ans.		*Correct diagram 2M*
	d) Ans.	**Define CPU bound program and I/O bound program** **CPU bound program**: If execution of a program is highly dependent on the CPU then it is known as CPU bound program. **OR** If a task does a lot operations using CPU, it's called a CPU-bound task. **I/O bound program**: If execution of a program is dependent on the input-output system and its resources, such as disk drives and peripheral devices then it is known as I/O bound program.	**2M** *Each definition 1M*
	e) Ans.	**Define paging and segmentation** **Paging** is a memory management scheme that permits the physical address space of process to be noncontiguous. Logical memory is divided into blocks of same size called as pages. **OR** **Paging** is a storage mechanism used in OS to retrieve processes from secondary storage to the main memory as pages. **OR** The process of retrieving processes in the form of pages from the secondary storage into the main memory is known as **paging**. **Segmentation** is a memory management scheme that permits dividing logical address space into multiple segments. **OR** **Segmentation** divides the Computer's physical memory and program's address space into segments.	**2M** *Correct definition 1M each*

	f) Ans.	**What is the use of PS command? Write long forms of UID, PID in the output of this command.** **ps command:** It is used to display the characteristics of a process. This command is used to list the processes associated with a user at a particular terminal. UID - (Owner) User-id PPID - Parent Process-id	**2M** ***Use of ps command 1M*** ***full form of each 1/2M***
	g) Ans.	**List any four file operations.** • Creating a file • Writing a file • Reading a file • Repositioning within a file • Deleting a file • Appending new information to the end of the file • Renaming an existing file. • Truncating a file • Creating copy of a file, copy file to another I/O device such as printer or display	**2M** ***Any four operations 1/2M each***
2.	 a) Ans.	**Attempt any <u>THREE</u> of the following:** **Describe multiprocessor OS with its advantages (any two)** • Multiprocessor systems are also known as parallel systems or tightly coupled systems. • These systems have two or more processors in close communication and they share computer resources such as bus, clock, memory and peripheral devices. • The whole task of multiprocessing is managed by the operating system, which allocates different tasks to be performed by the various processors in the system. • Applications designed for the use in multiprocessing are said to be threaded, which means that they are broken into smaller routines that can be run independently. • Multiple CPUs are linked together so that a job can be divided and executed more quickly. When a job is completed, the results from all CPUs are compiled to provide the final output.	**12** **4M** ***Description of multiprocessor OS-3M*** ***Any two advantages 1/2M each***



OR



Note: ***Any other relevant advantage shall be considered***

Advantages

- **Increased reliability:** Due to the multiprocessing system, processing tasks can be distributed among several processors. This increases reliability as if one processor fails; the task can be given to another processor for completion.
- **Increased throughout:** As several processors increase, more work can be done in less time.
- **The economy of Scale:** As multiprocessors systems share peripherals, secondary storage devices, and power supplies, they are relatively cheaper than single-processor systems.
- **Reduce Cost**

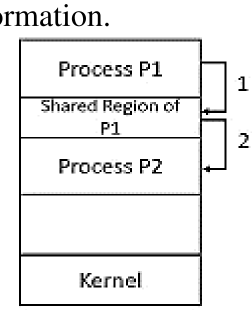
b) **Write down the responsibilities of the following components of OS** — **4M**

i) **Memory Management**
ii) **File Management**

Responsibilities of each management 2M

Ans. **i) Memory Management:**

- Keeping track of which parts of memory are currently being used and by whom.

		• Deciding which processes (or parts thereof) and data to move into and out of memory. • Allocating and deallocating memory space as needed. • Managing swap spaces, which store inactive pages of memory. • Implementing policies for memory allocation. **File Management:** • The creation and deletion of files. • The creation and deletion of directory • The support of primitives for manipulating files and directions. • The mapping of files onto secondary storage. • The backup of files on stable storage media. • Adding and editing the data in files. • Moving files from one location to another. • Store, arrange, or accessing files on a disk or other storage locations.	
	c) **Ans.**	**Explain shared memory method of IPC using neat labelled diagram** **Inter-process communication:** Cooperating processes require an Inter- process communication (IPC) mechanism that will allow them to exchange data and information.  Shared Memory System • In this, all processes who want to communicate with other processes can access a region of the memory residing in an address space of a process creating a shared memory segment. • All the processes using the shared memory segment should attach to the address space of the shared memory. All the processes can exchange information by reading and/or writing data in shared memory segment.	**4M** ***Explanation* 3M, *diagram* 1M**

		• The form of data and location are determined by these processes who want to communicate with each other. • These processes are not under the control of the operating system. • The processes are also responsible for ensuring that they are not writing to the same location simultaneously. • After establishing shared memory segment, all accesses to the shared memory segment are treated as routine memory access and without assistance of kernel.	
	d) Ans.	**Explain following terms with respect to scheduling** **i) CPU utilization** **ii) Throughput** **iii) Turnaround time** **iv) Waiting time** • **CPU utilization:** In multiprogramming the main objective is to keep CPU as busy as possible. CPU utilization can range from 0 to 100 percent. • **Throughput:** It is the number of processes that are completed per unit time. It is a measure of work done in the system.Throughput depends on the execution time required for any process. • **Turnaround time:** The time interval from the time of submission of a process to the time of completion of that process is called as turnaround time. It is the sum of time period spent waiting to get into the memory, waiting in the ready queue, executing with the CPU, and doing I/O operations. • **Waiting time:** It is the sum of time periods spent in the ready queue by a process. When a process is selected from job pool, it is loaded into the main memory. A process waits in ready queue till CPU is allocated to it.	**4M** ***Each term 1M***
3.	a)	**Attempt any <u>THREE</u> of the following:** **Explain following commands with their syntax** **i) kill** **ii) Sleep** **iii) Wait** **iv) Exit**	**12** **4M**

	Ans.	**i)kill** **Syntax: kill pid** Kill command is used to stop execution of particular process by sending an interrupt signal to the process **ii)Sleep** **Syntax: sleep NUMBER[SUFFIX]** The sleep command pauses the execution for specified time in command. **iii)Wait** **Syntax: wait [pid]** Wait command waits for running process to complete and return the exit status. **iv) Exit** **Syntax: exit** used to quit the shell **(OR)** **Syntax: exit[n]** The terminal window will close and return a status of n	***Each command Syntax ½M*** ***Explanation ½ M***
	b) **Ans.**	**What is deadlock? Discuss any one method of deadlock prevention** A deadlock is a situation where a set of processes are blocked because each process is holding a resource and waiting for another resource acquired by some other process. Methods for deadlock prevention (***Any 1)*** **1. Eliminate Mutual Exclusion:** The mutual-exclusion condition must hold for non-sharable resources. Mutual section from the resource point of view is the fact that a resource can never be used by more than one process simultaneously which is fair enough but that is the main reason behind the deadlock. If a resource could have been used by more than one process at the same time then the process would have never been waiting for any resource.Read-only files are a good example of a sharable resource. If several processes attempt to open a read-only file at the same time, they can be granted simultaneous access to the file.	**4M** ***Deadlock definition 1M*** ***Any 1 method 3M***

		A process never needs to wait for a sharable resource.Sharable resources, in contrast, do not require mutually exclusive access and thus cannot be involved in a deadlock **2. Eliminate Hold and Wait:** One way to avoid this Hold and Wait is when a process requests a resource; it does not hold any other resources. One protocol that can be used requires each process to request and be allocated all its resources before it begins execution. Another protocol that can be used is, to allow a process to request resources only when the process has none. A process may request some resources and use them. Before it requests any additional resources, it must release all the resources that are currently allocated to it. **3. Eliminate No Preemption:** If a process that is holding some resources requests another resource that cannot be immediately allocated to it, then all resources currently being held are preempted. That is this resources are implicitly released. The preempted resources are added to the list of resources for which the process is waiting. Process will be restarted only when all the resources i.e. its old resources, as well as the new ones that it is requesting will be available.Preemption ensures that resources are efficiently utilized and prevents deadlocks caused by the hold and wait condition. **4. Eliminate Circular Wait:** Circular-wait condition never holds is to impose a total ordering of all resource types, and to require that each process requests resources in an increasing order of enumeration. Let R = {R1, R2, ..., Rn} be the set of resource types. We assign to each resource type a unique integer number, which allows us to compare two resources and to determine whether one precedes another in our ordering. Formally, define a one-to-one function F: R _ N, where N is the set of natural numbers.	
	c)	**Describe concept of free space management technique using bit map method**	4M

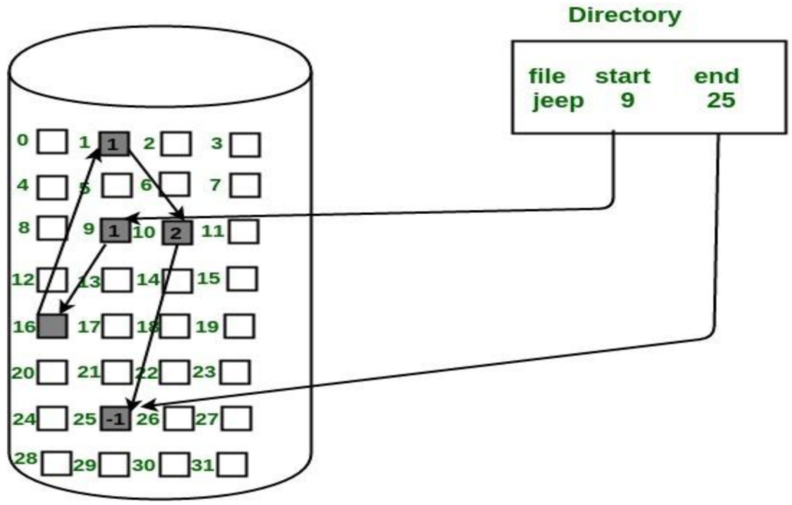
	Ans.	**Bit Map:** The free-space list is implemented as a bit map or bit vector. Each block is represented by 1 bit. If the block is free, the bit is 1; if the block is allocated, the bit is 0. For example, consider a disk where blocks 2, 3, 4, 5, 8, 9, 10, 11, 12, 13 are free and the rest of the blocks are allocated. The free-space bit map would be : 0011110011111100	*Correct Explanation 4M*

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	0	1	1	1	1	0	0	1	1	1	1	1	1	0	0

1=Free block 0=Allocated block

The main advantage of this approach is its relative simplicity and its efficiency in findingthe first free block or n consecutive free blocks on the disk.

	d) Ans.	**Draw the diagram of linked file allocation method and explain it.** **Linked Allocation:** This allocation is on the basis of an individual block. Each block contains a pointer to the next block in the chain. • In this scheme, each file is a linked list of disk blocks which need not be contiguous. The disk blocks can be scattered anywhere on the disk. • The directory contains a pointer to the first and the last blocks of the file. • Creating a file is easy. We simply create a new entry in the device directory. A write to a file removes first free block from free space list and write to it. This new block is then linked to the end of the file. To read a file, we simply read block by following the pointers from block to block. • There is no external fragmentation with linked allocation.	**4M** *Correct Diagram 2M* *Explanation 2M*

		• Space is required for the pointers, 1.5% of disk is used for the pointers and not for information. If a pointer is lost or damaged or bug occurs in operating system or disk hardware failure occur, it may result in picking up the wrong pointer. • This method cannot support direct access.  • This method is used only for a sequential access files • This method requires more space to store pointers • So instead of blocks, clusters are used for allocation but this creates internal fragmentation.	
4.	**a)**	**Attempt any <u>THREE</u> of the following:** **Compare between CLI based OS and GUI based OS (Any four points)**	**12** **4M**

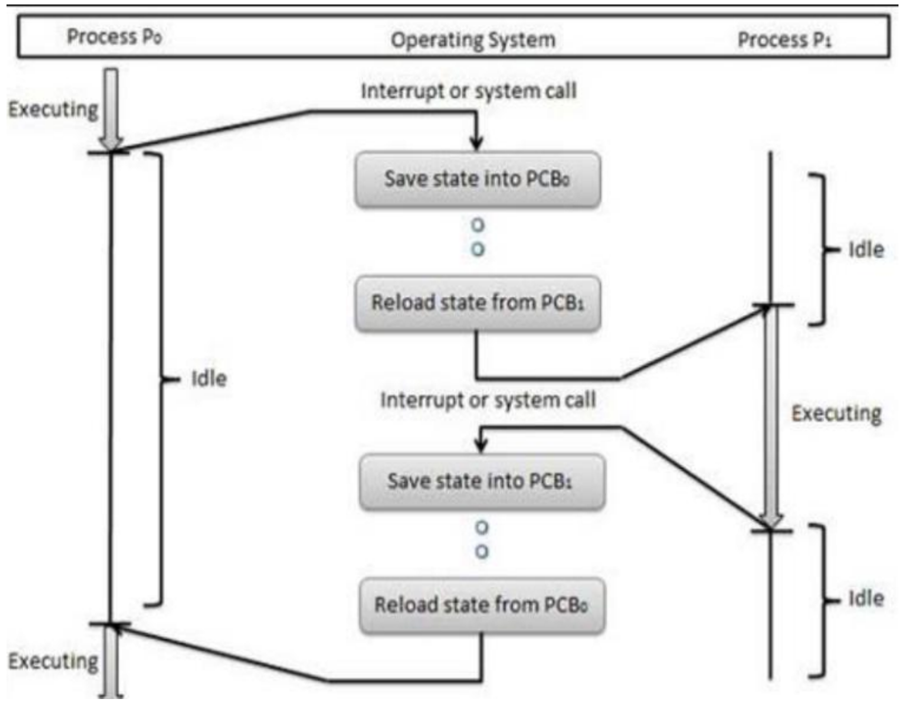
WINTER – 2023 EXAMINATION
MODEL ANSWER

Ans.

	COMMAND LINE INTERFACE (CLI)	GRAPHIC USER INTERFACE (GUI)
Definition	Interaction is by typing commands	Interaction with devices is by graphics and visual components and icons
Understanding	Commands need to be memorized	Visual indicators and icons are easy to understand
Memory	Less memory is required for storage	More memory is required as visual components are involved.
Working Speed	Use of keyboard for commands makes CLI quicker and faster	Use of mouse for interaction makes it slow
Resources used	Only keyboard	Mouse and keyboard both can be used
Accuracy	High	Comparatively low
Menu	In CLI, there are no menus provided.	While in GUI, menus are provided
Flexibility	Command line interface does not change, remains same over time	Structure and design can change with updates
Example	DOS, UNIX	Windows, Linux

Any 4 points 1M each

	b) Ans.	**What are the different types of system calls? Give examples of each** Types of System Calls: 1.Process Control 2.File management 3.Device Management 4.Information Maintenance 5.Communication: **1. System calls for Process Control:** ● Create/ Terminate process ● Load/Execute Process. ● End/Abort process. ● Ready process/Dispatch process. ● Suspend process/Resume Process. ● Get/Set Process Attributes. ● Wait event, signal event ● Allocate and deallocate memory **2. System call for File management:** ● create new file, delete existing file ● open, close file ● Create and delete directories ● read, write, reposition ● get/set file attributes **3. System call for Device Management:** ● request device, release device ● read, write, reposition ● get/set device attributes ● logically attach or detach devices **4. System call for Information Maintenance:** ● get/set time or date ● get/set system data ● get/set process, file, or device attributes **5. System Communication:** ● create, delete communication connection ● send, receive messages ● transfer status information ● attach or detach remote devices	**4M** ***Listing of Types 2M*** ***Example of any two system calls 2M***

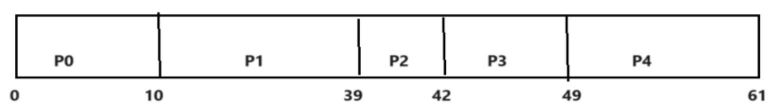
	c)	**Explain working of CPU switch from process to process with neat labelled diagram.**	**4M**
	Ans.	A CPU switch from process to process is referred as **context switch**. A context switch is a mechanism that store and restore the state or context of a CPU in Process Control block sothat a process execution can be resumed from the same point at a later time. When thescheduler switches the CPU from one process to another process, the context switch savesthe contents of all process registers for the process being removed from the CPU, in its process control block. Context switch includes two operations such as state save and state restore. State save operation stores the current information of running process into its PCB. State restoreoperation restores the information of process to be executed from its PCB. Switching the CPU from one process to another process requires performing state save operation for thecurrently executing process (blocked) and a state restore operation for the process ready for execution. This task is known as context switch.  	***Explanation 2M*** ***Diagram 2M*** ***Relevant Explanation shall be considered***

d) Solve given problem by using FCFS scheduling algorithm. Draw correct Gantt chart and calculate average waiting time and average turnaround time ***4M***

Process	Arrival Time	Burst Time
P0	0	10
P1	1	29
P2	2	3
P3	3	7
P4	4	12

Ans.

Gantt Chart



Gantt chart 2M,

FCFS

Waiting time
P0=0
P1= (10-1) = 9
P2= (39-2) = 37
P3= (42-3) = 39
P4= (49-4) = 45

Average waiting time calculation 1M,

Average Waiting time = Waiting time of all processes / Number of processes
= (0+9+37+39+45)/5 = 26 ms

Turnaround Time
P0=10
P1=39-1=38
P2=42-2=40
P3=49-3=46
P4=61-4=57

Average turnaround time calculation 1M

Average Turnaround time = Turnaround time of all processes / Number of processes
= (10+38+40+46+57)/5
= 191/5
= 38.2 ms

e) **Which hole is taken for next segment request for 8 KB in a swapping system for First fit, Best fit and Worst fit.** *4M*

OS
4 KB
9 KB
20 KB
16 KB
8 KB
2 KB
6 KB

Ans. *Any Correct representation 4M*

First Fit: Allocate the first free block to the new process

OS
4 KB
<FREE> 1 KB 8 KB
20 KB
16 KB
8 KB
2 KB
6 KB

Best Fit: Allocate the smallest free block that is big enough to accommodate new process

OS
4 KB
9 KB
20 KB
16 KB
<FREE> 0 KB 8 KB
2 KB
6 KB

Worst fit: Allocate the largest free block to the new process

OS
4 KB
9 KB
<FREE> 12 KB 8 KB
16 KB
8 KB
2 KB
6 KB

OR

First Fit: Allocate the first free block to the new process

OS
4 KB
9 KB
20 KB
16 KB
8 KB
2 KB
6 KB

Best Fit: Allocate the smallest free block that is big enough to accommodate new process

OS
4 KB
9 KB
20 KB
16 KB
8 KB
2 KB
6 KB

Worst fit: Allocate the largest free block to the new process

OS
4 KB
9 KB
20 KB
16 KB
8 KB
2 KB
6 KB

WINTER – 2023 EXAMINATION
MODEL ANSWER

5.		**Attempt any <u>TWO</u> of the following:**	**12**
	a)	**Write two uses of the following operating system tools:** **i) Security Policy** **ii) User Management** **iii) Performance Monitor**	**6M**
	Ans.	**<u>i) Security Policy:</u>** -As OS security policies and procedures cover a large area, there are following techniques to addressing them: • Installing and updating anti-virus software • Ensure the systems are patched or updated regularly $ yum updates $yum check-update • Implementing user management policies to protect user accounts and privileges. • Installing a firewall and ensuring that it is properly set to monitor all incoming and outgoing traffic. - **Use Secure SSH:** Telnet and rlogin protocols uses plain text, not encrypted format which is the security breaches. SSH is a secure protocol that use encryption technology during communication with server. Never login directly as root unless necessary. Use “sudo” to execute commands. $ vi /etc/ssh/sshd_config $ PermitRootLogin no #disable root login $AllowUsers username # Allow only specific users $ Protocol 2 #use SSH protocol 2 version **Lock and Unlock Features:** They are very useful, instead of removing an account from the system, you can lock it for an week or a month. To lock a specific user, you can use the follow command. $ passwd -1 accountname **Turn Off IPv6:** If you are not using a IPv6 protocol, then you should disable it because most of the applications or policies not required IPv6 protocol and currently it doesn’t required on the server. Go to network configuration file and add followings lines to disable it. # vi /etc/sysconfig/network NETWORKING_IPV6=no IPV6INIT=no	***Each tool with any 2 valid uses 2M***

Enables Iptables (Firewall): It is highly recommended to enable Linux firewall to secure unauthorised access of your servers. Apply rules in iptables to filters incoming, outgoing and forwarding packets. We can specify the source and destination address to allow and deny in specific udp/tcp port number.

Use Strong Password Policy: Passwords are always a security problem because humans are. People often cannot be bothered to come up with a lot of different passwords, so they use the same ones in different places or combinations that are easy to remember, like "password" or "abcde". Basically, a gift to hackers. Make it a requirement that any password must contain both upper and lower case, be a mix of numbers, letters and symbols and you'll be way safer.

ii) User Management:

-As the administrator, it is your job to create and manage the accounts for all required users.

-User management includes everything from creating a user to deleting a user on your system and also assigning the passwords to created users using "passwd" command.

Following are the Linux command line tools for managing users and groups:

1. useradd.
2. usermod.
3. userdel.

iii) Performance Monitor

It is really very tough job for every System or Network administrator to monitor and debug Linux System Performance problems every day.

- The **top** command used to display all the running and active real-time processes in ordered list and updates it regularly. It display CPU usage, Memory usage, Swap Memory, Cache Size, Buffer Size, Process PID, User, Commands and much more. It also shows high memory and CPU utilization of a running processes.
- Linux **vmstat** command used to display statistics of virtual memory, kernel threads, disks, system processes, I/O blocks, interrupts, CPU activity and much more.

	b) Ans.	**Differentiate between process and thread (any two Points). Also discuss the benefits of multithreaded programming.**	**6M**

	Process	Thread
Definition	A process is a program under execution i.e. an active program.	A thread is a lightweight process that can be managed independently by a scheduler.
Running mechanism	Processes run in separate memory spaces.	Threads within the same process run in a shared memory space.
Concept	Process is heavy weight and any program is in execution.	It is a lightweight process and a segment of a process.
PCB	The process has its own Process Control Block, Stack, and Address Space.	Thread has Parents' PCB, its own Thread Control Block, and Stack and common Address space.
Context Switching	Context switching between the process is more expensive	Context switching between threads of the same process is less expensive
Dependency	Processes are independent.	Threads are dependent
Controlling	Process is controlled by the operating system.	Threads are controlled by programmer in a program

Differentiate - any 2 valid points 2M

Benefits 4M

Benefits of multithreaded programming:

The benefits of multithreaded programming can be broken down into four major categories:

Responsiveness:

Multithreading an interactive application may allow a program to continue running even if part of it is blocked or is performing a lengthy operation, thereby increasing responsiveness to the user. For

		instance, a multithreaded Web browser could allow user interaction in one thread while an image was being loaded in another thread. **Resource sharing:** Processes may only share resources through techniques such as shared memory or message passing. Such techniques must be explicitly arranged by the programmer. However, threads share the memory and the resources of the process to which they belong by default. The benefit of sharing code and data is that it allows an application to have several different threads of activity within the same address space. **Economy:** Allocating memory and resources for process creation is costly. Because threads share the resources of the process to which they belong, it is more economical to create and context-switch threads. Empirically gauging the difference in overhead can be difficult, but in general it is much more time consuming to create and manage processes than threads. In Solaris, for example, creating a process is about thirty times slower than is creating a thread, and context switching is about five times slower. **Scalability:** The benefits of multithreading can be greatly increased in a multiprocessor architecture, where threads may be running in parallel on different processors. A single-threaded process can only run on one processor, regardless how many are available. Multithreading on a multi-CPU machine increases parallelism.	
	c)	**Find out the total number of page faults using:** **i. Least Recently used Page Replacement** **ii. Optimal Page Replacement** **Page replacement algorithms of memory management, if the pages are coming in the order:** **7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1**	**6M**

Ans

Any other frame size shall be considered. Calculation of page faults depends on frame size.

i) LRU: *LRU 3M*

Considering frame size is 3:

Ref	7	0	1	2	0	3	0	4	2	3	0	3	2	1	2	0	1	7	0	1
F1	7	7	7	2	2	2	2	4	4	4	0	0	0	1	1	1	1	1	1	1
F2		0	0	0	0	0	0	0	0	3	3	3	3	3	3	0	0	0	0	0
F3			1	1	1	3	3	3	2	2	2	2	2	2	2	2	2	7	7	7
Fault	F	F	F	F		F		F	F	F	F			F		F		F		

Total Page faults using LRU Page Replacement/Fault=12

ii) Optimal: *Optimal 3M*

Considering Frame size is 3:

Ref	7	0	1	2	0	3	0	4	2	3	0	3	2	1	2	0	1	7	0	1
F1	7	7	7	2	2	2	2	2	2	2	2	2	2	2	2	2	2	7	7	7
F2		0	0	0	0	0	0	4	4	4	0	0	0	0	0	0	0	0	0	0
F3			1	1	1	3	3	3	3	3	3	3	3	1	1	1	1	1	1	1
Fault	F	F	F	F		F		F			F			F				F		

Total Page faults using Optimal Page Replacement/Page Fault=09

6. **Attempt any TWO of the following:** **12**

a) **How pre-emptive scheduling is better than non pre-emptive scheduling by solving following problem using SJF (Solve it by using pre-emptive SJF and non-pre-emptive SJF also).** **6M**

Process	Arrival time	Burst time
P1	0	8
P2	1	4
P3	2	9
P4	3	5

Ans.

Following are the reason, why pre-emptive scheduling is better than non pre-emptive scheduling:

Reason 2M

Pre-emptive SJF 2M

NonPre-emptive SJF 2M

- Preemptive scheduling is quite flexible because critical processes are allowed to access the CPU because they come in the ready queue and no matter which process is currently running. Non-preemptive scheduling is tough because if an essential process is assigned to the ready queue, the CPU process is not be interrupted.

- Preemptive scheduling allows a process to be interrupted in the midst of its execution, taking the CPU away and allocating it to another process. Non-preemptive scheduling ensures that a process relinquishes control of the CPU only when it finishes with its current CPU burst.

* Pre-emptive SJF :-

Process	Arrival Time	Burst Time	Waiting Time (ms)	Turn - Around Time (ms)
P1	0	8	10-1 = 9	17-0 = 17
P2	1	4	1-1 = 0	5-1 = 4
P3	2	9	17-2 = 15	26-2 = 24
P4	3	5	5-3 = 2	10-3 = 7

- Gantt Chart :-

P1	P2	P4	P1	P3
0	1	5	10	17

(end: 26)

- Average Waiting Time = $\frac{9+0+15+2}{4} = \frac{26}{4} = \boxed{6.5\text{ ms}}$

- Average Turn-Around Time = $\frac{17+4+24+7}{4} = \frac{52}{4} = \boxed{13\text{ms}}$

* Non-Pre-emptive SJF :-

Process	Arrival Time	Burst Time	Waiting Time (ms)	Turn-Around Time (ms)
P1	0	8	0	8-0 = 8
P2	1	4	8-1 = 7	12-1 = 11
P3	2	9	17-2 = 15	26-2 = 24
P4	3	5	12-3 = 9	17-3 = 14

- Gantt chart :-

P1	P2	P4	P3
0	8	12	17

(end: 26)

- Average Waiting Time = $\frac{0+7+15+9}{4} = \frac{31}{4} = \boxed{7.75\text{ ms}}$

- Average Turn-Around Time = $\frac{8+11+24+14}{4} = \frac{57}{4} = \boxed{14.25\text{ms}}$

b) **List free space management technique with the help of neat diagram, explain any one technique in detail.** **6M**

Listing free space management 1M

Explanation 3M

Diagram 2M

Ans. A file system is responsible to allocate the free blocks to the file therefore it has to keep track of all the free blocks present in the disk. There are mainly two approaches by using which, the free blocks in the disk are managed.

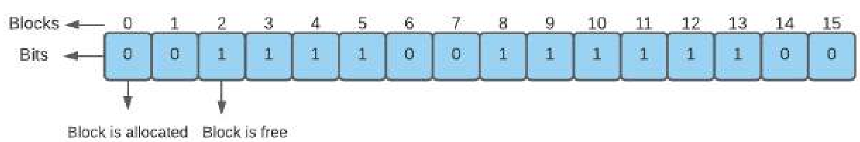
Following are the free space management technique:

1) Bitmap
2) Linked List
3) Grouping

Bit Vector

The first method that we will discuss is the bit vector method. Also known as the bit map, this is the most frequently used method to implement the free space list. In this method, each block in the hard disk is represented by a bit (either 0 or 1). If a block has a bit 0 means that the block is allocated to a file, and if a block has a bit 1 means that the block is not allocated to any file, i.e., the block is free.

For example, consider a disk having 16 blocks where block numbers 2, 3, 4, 5, 8, 9, 10, 11, 12, and 13 are free, and the rest of the blocks, i.e., block numbers 0, 1, 6, 7, 14 and 15 are allocated to some files. The bit vector for this disk will look like this-



We can find the free block number from the bit vector using the following method-

Block number = (Number of bits per word)* (number of 0-value words) + (offset of first bit)

We will now find the first free block number in the above example.

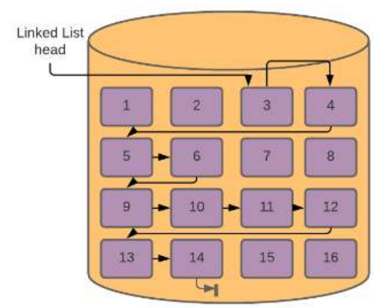
The first group of 8 bits (00111100) constitutes a non-zero word since all bits are not 0. After finding the non-zero word, we will look for the first 1 bit. This is the third character of the non-zero word. Hence, offset = 3.

Therefore, the first free block number = 8 * 0 + 3 = 3.

Linked List:

Another method of doing free space management in operating systems is a linked list. In this method, all the free blocks existing in the disk are linked together in a linked list. The address of the first

free block is stored somewhere in the memory. Each free block contains a pointer that contains the address to the next free block. The last free block points to null, indicating the end of the linked list.
For example, consider a disk having 16 blocks where block numbers 3, 4, 5, 6, 9, 10, 11, 12, 13, and 14 are free, and the rest of the blocks, i.e., block numbers 1, 2, 7, 8, 15 and 16 are allocated to some files. If we maintain a linked list, then Block 3 will contain a pointer to Block 4, and Block 4 will contain a pointer to Block 5.



Grouping: The third method of free space management in operating systems is grouping. This method is the modification of the linked list method. In this method, the first free block stores the addresses of the n free blocks. The first n-1 of these blocks is free. The last block in these n free blocks contains the addresses of the next n free blocks, and so on.
For example, consider a disk having 16 blocks where block numbers 3, 4, 5, 6, 9, 10, 11, 12, 13, and 14 are free, and the rest of the blocks, i.e., block numbers 1, 2, 7, 8, 15 and 16 are allocated to some files.
If we apply the Grouping method considering n to be 3, Block 3 will store the addresses of Block 4, Block 5, and Block 6. Similarly, Block 6 will store the addresses of Block 9, Block 10, and Block 11. Block 11 will store the addresses of Block 12, Block 13, and Block 14. This is also represented in the following figure

Block 3 -> 4, 5, 6
Block 6 -> 9, 10, 11
Block 11 -> 12, 13, 14

c) **Draw and explain directory structure of a file system in terms of single level, two level and tree structure.** **6M**

Ans.

A **directory** is a container that is used to contain folders and files.
Following are the logical structures of a directory:
1) Single level
2) Two level
3) Tree structure

Relevant explanation shall be considered

Single level Diagram 1M Explanation 1M

Two Level Diagram 1M Explanation 1M

Tree Level Diagram 1M

Explanation 1M

Single-level directory:

The single-level directory is the **simplest directory structure**.
In it, all files are contained in the same directory, which makes it easy to support and understand.
Single level directory has a significant limitation, however, when the number of files increases or when the system has more than one user. Since all the files are in the same directory, they must have a **unique name**. If two users call their dataset test, then the unique name rule violated.

- •Simple operations like file creation, search, deletion, and updating are possible with a single-level directory structure.
- •The single-level directory is easier to understand in practical life.
- •Two file names cannot be the same. In case two files are given the same name, the previous one is overridden.
- •If the number of files is very large, searching a particular file is very inefficient.
- •Segregation of important and unimportant files is not possible.
- •The single-level directory is not useful for multi-user systems.

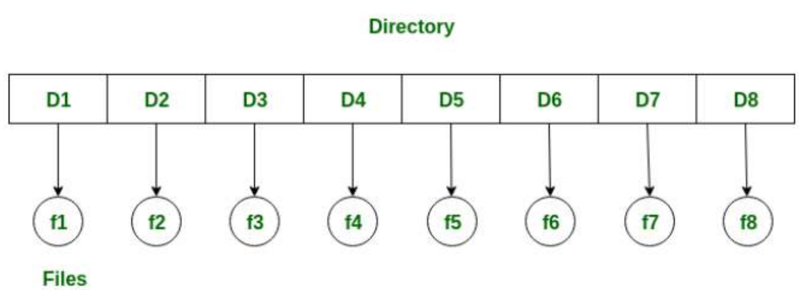


Fig:Single level Directory Structure

Two-level directory:

As we have seen, a single level directory often leads to confusion of files names among different users. The solution to this problem is to create a **separate directory for each user**.

In the two-level directory structure, each user has their own **user files directory (UFD).** The UFDs have similar structures, but each lists only the files of a single user. System's **master file directory (MFD)** is searched whenever a new user id is created.

- Searching is very easy.
- There can be two files with the same name in two different user directories. Since they are not in the same directory, the same name can be used.
- Grouping is easier.
- A user cannot enter another user's directory without permission.
- Implementation is easy.
- It does not allow users to create subdirectories.

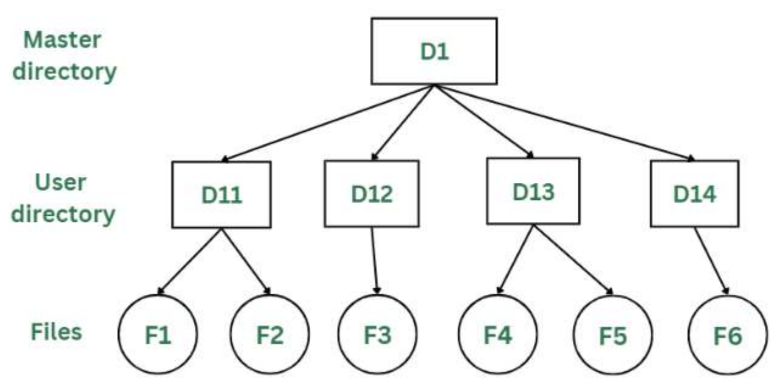


Fig: Two Level Directory structure

Tree Structure/ Hierarchical Structure:

Tree directory structure of operating system is most commonly used in our **personal computers**. User can create files and subdirectories too, which was a disadvantage in the previous directory structures.

This directory structure resembles a real tree upside down, where the **root directory** is at the peak. This root contains all the directories for each user. The users can create subdirectories and even store files in their directory.

A user do not have access to the root directory data and cannot modify it. And, even in this directory the user do not have access to other user's directories. The structure of tree directory is given below which shows how there are files and subdirectories in each user's directory.

•The root directory is highly secured, and only the system administrator can access it. We can see how there can be subdirectories inside the user directories. A user cannot modify the root directory data. Also, a user cannot access another user's directory.
•Allows subdirectories inside a directory.
•Searching is easy.
•Allows grouping.
•Segregation of important and unimportant files is easy.
•As one user cannot enter another user's directory, this restricts sharing of files.
•Too many subdirectories may make the search complicated.
•Users cannot modify the root directory's data.

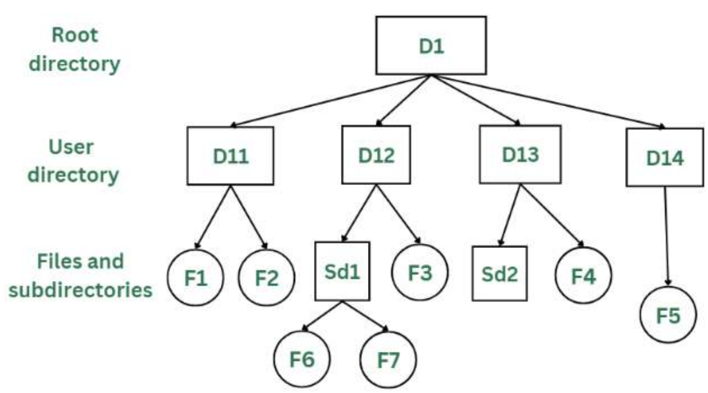


Fig: Tree Structure/ Hierarchical Structure